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Faculty of Computer Science & Engineering

Robustness of Graph-Based Fraud Detection Under Camouflage and Graph Shift

A controlled benchmark on YelpChi

Conceptual illustration

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Social Networks and Media

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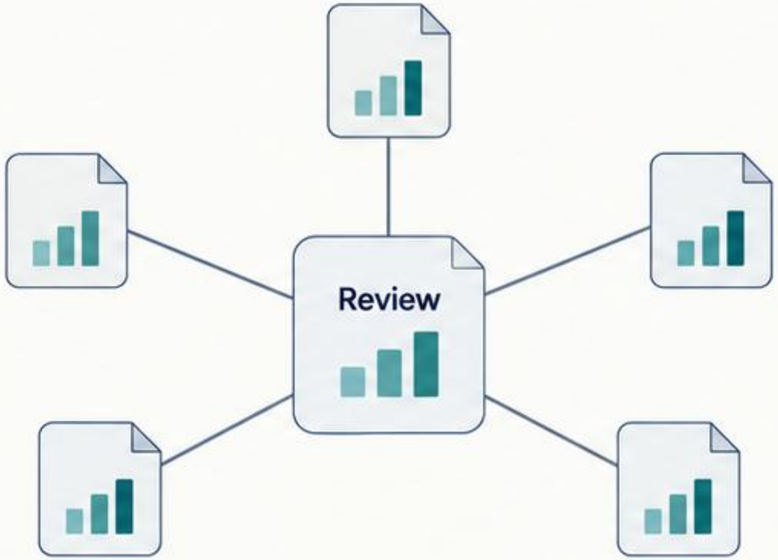
Skopje, 2026

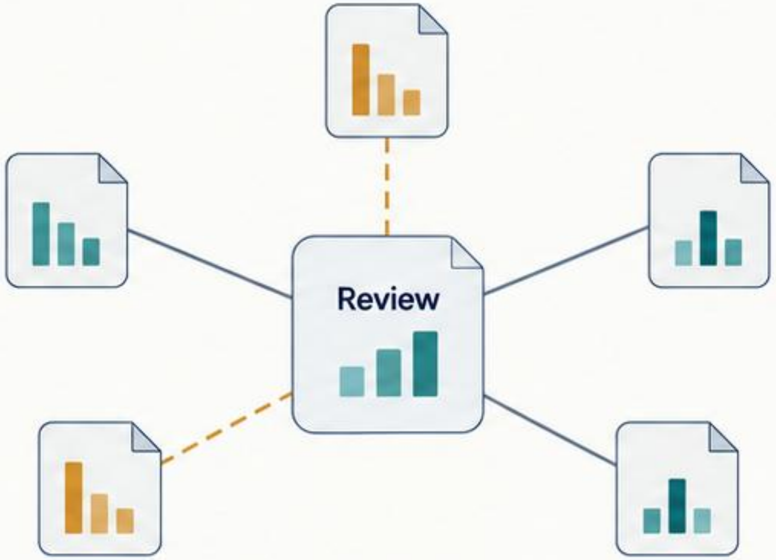
Relationships add evidence—and a dependency

Useful neighborhood context



Relevant neighbor information

Misleading neighborhood context



Noisy or altered neighbor information

A graph model combines a review's **own features** with information from its **neighbors**.

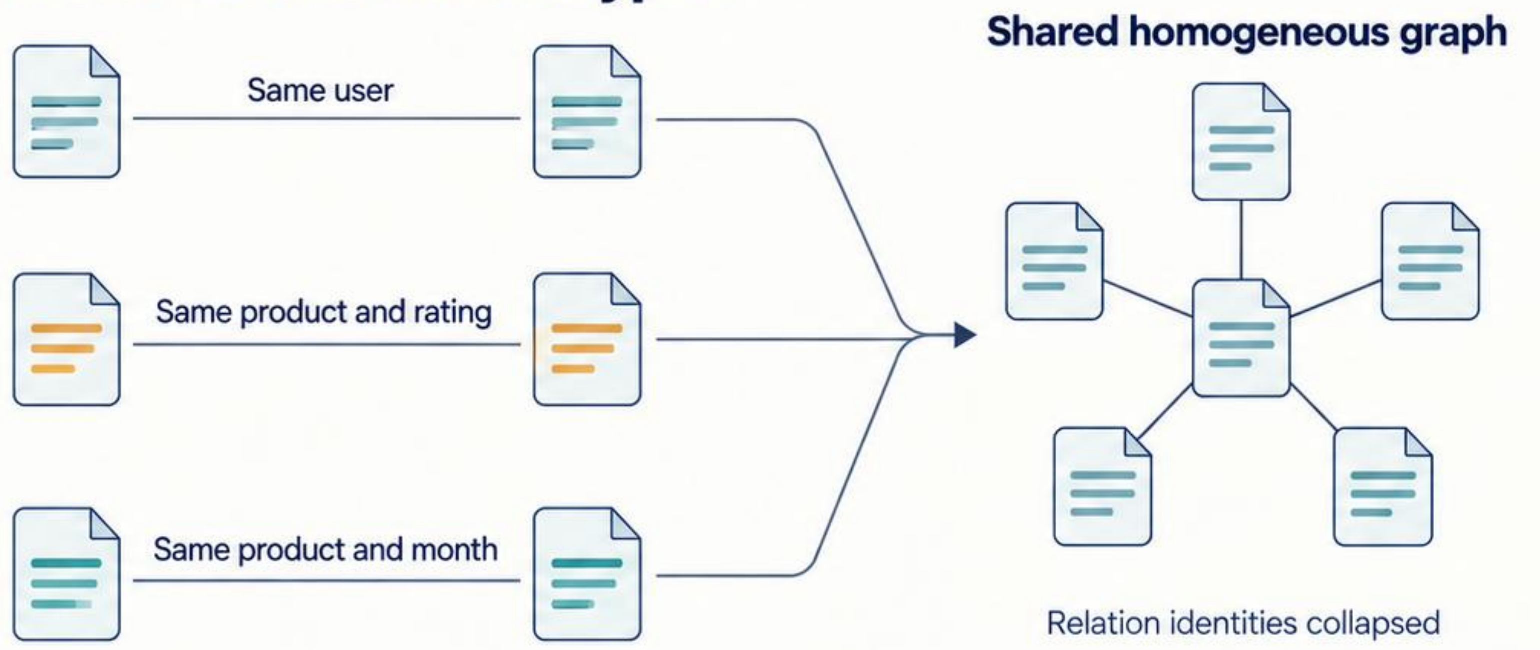
A — A **Homophily**: same-label neighbors

A — B **Heterophily**: different-label neighbors

Different labels do not automatically mean harmful information.

YelpChi represents reviews as a graph

Three source relation types



45,954
review nodes

32
features per review

8,051,348
clean directed edge entries

14.53%
positive reviews
6,677 of 45,954

Average Precision is the primary metric.

Random-ranking reference: AP ≈ 0.145

Positive labels are Yelp-filtered reviews, not independently verified criminal behavior.

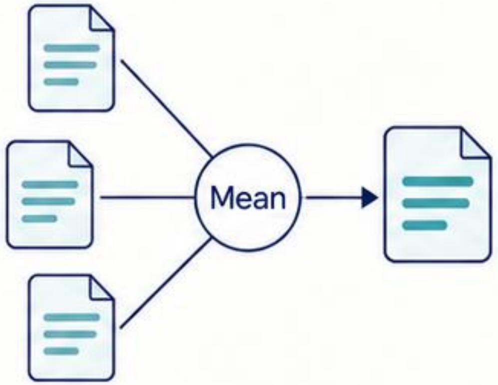
Four models test different uses of neighborhood information

MLP

Own features only

Role:
Graph-blind control

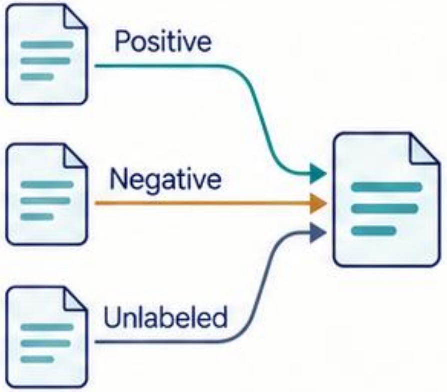
GraphSAGE



Mean neighbor aggregation

Role:
Conventional GNN baseline

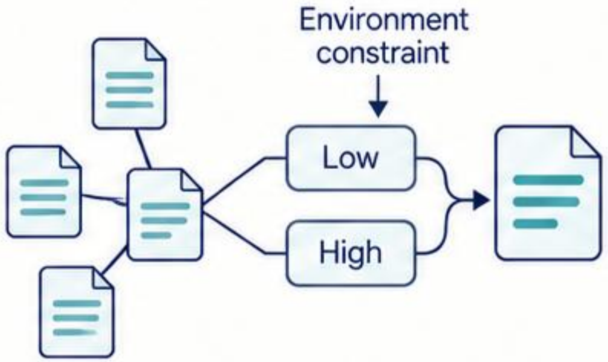
PMP adapter



Partitioned message passing

Role:
Specialized fraud model

SEC-GFD adapter



Spectral filtering + environmental constraint

Role:
Heterophily-aware model

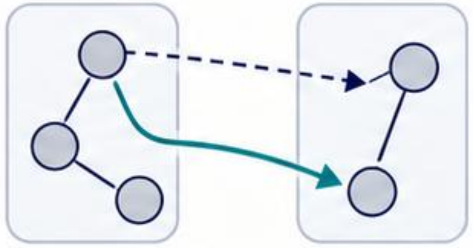
Shared data and evaluation. Different ways of using the graph.

PMP and SEC-GFD use feasibility adapters, not full paper reproductions.

Five stress families answer different questions

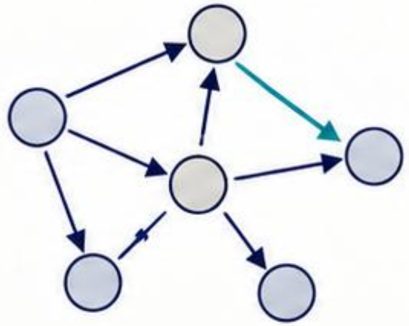
Non-oracle controlled stress

Construction does not use true labels



Feature-partition rewiring

Redirect edges across feature-derived groups.

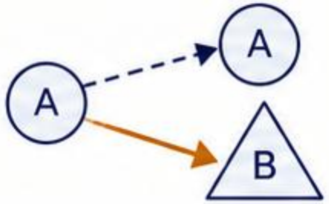


Uniform edge noise

Add random directed edges.

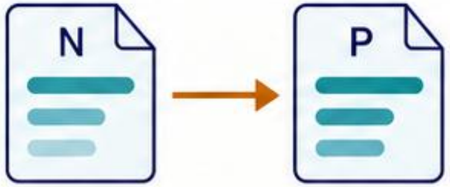
Oracle-assisted diagnostics

Construction uses true labels, including test labels



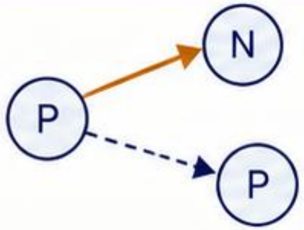
True-label rewiring

Redirect edges to opposite-label nodes.



Feature camouflage

Copy negative-node features onto selected positives.



Relation camouflage

Add negative neighbors and remove positive links.

Only the two non-oracle scenarios enter the operational ranking.

Severity is scenario-specific. Feature camouflage fully replaces each selected feature vector.

The same changed graph supports two different evaluations

Unexpected shift

`train_clean_eval_all`

Known changed environment

`train_on_variant`

Training exposure is part of the result.

Epoch and threshold selection use validation data, never test outcomes.

The benchmark separates three sources of variation

Two train / validation / test allocations | 40 / 20 / 40 | 60 / 20 / 20

Split assignment

3

split seeds
per allocation

Graph realization

2

graph seeds
per positive severity

Training randomness

5

training seeds

4 models · 2 protocols · 5 stress families

5,040
successful evaluations

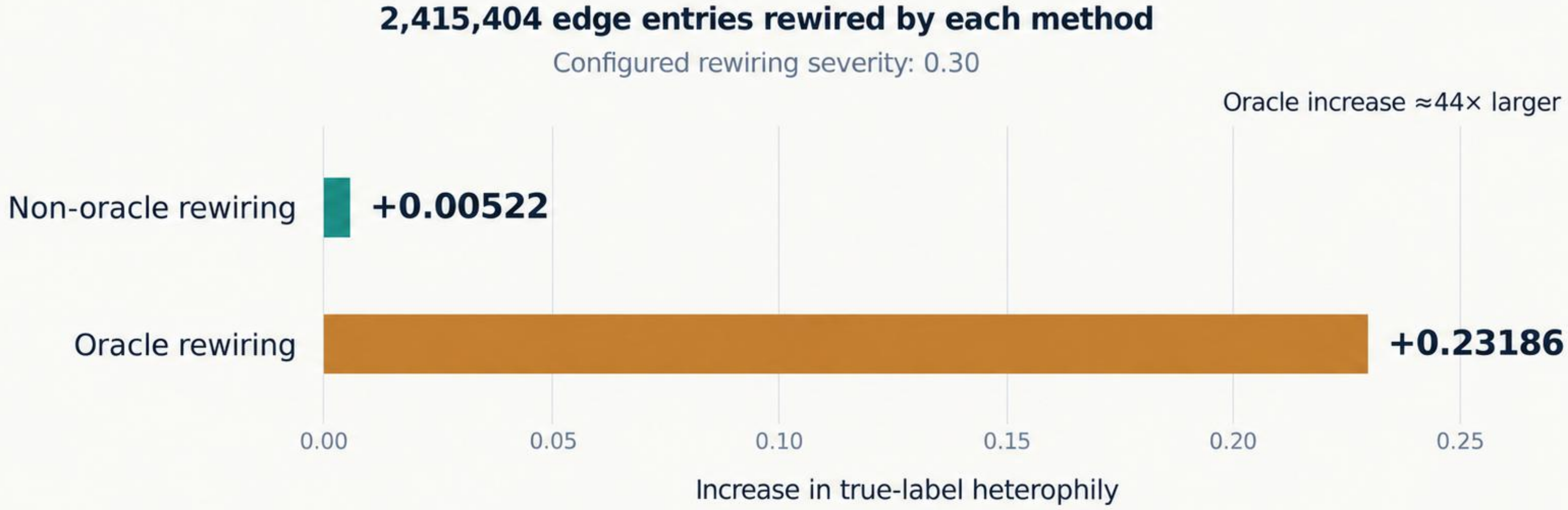
2,640
model fits

Clean-trained models are reused across stress evaluations.

Repeated runs test whether conclusions survive different sources of randomness.

Clean runs have no independent perturbation seed. Split-level intervals are descriptive (3 splits).

Equal rewiring budgets produced very different label mixing



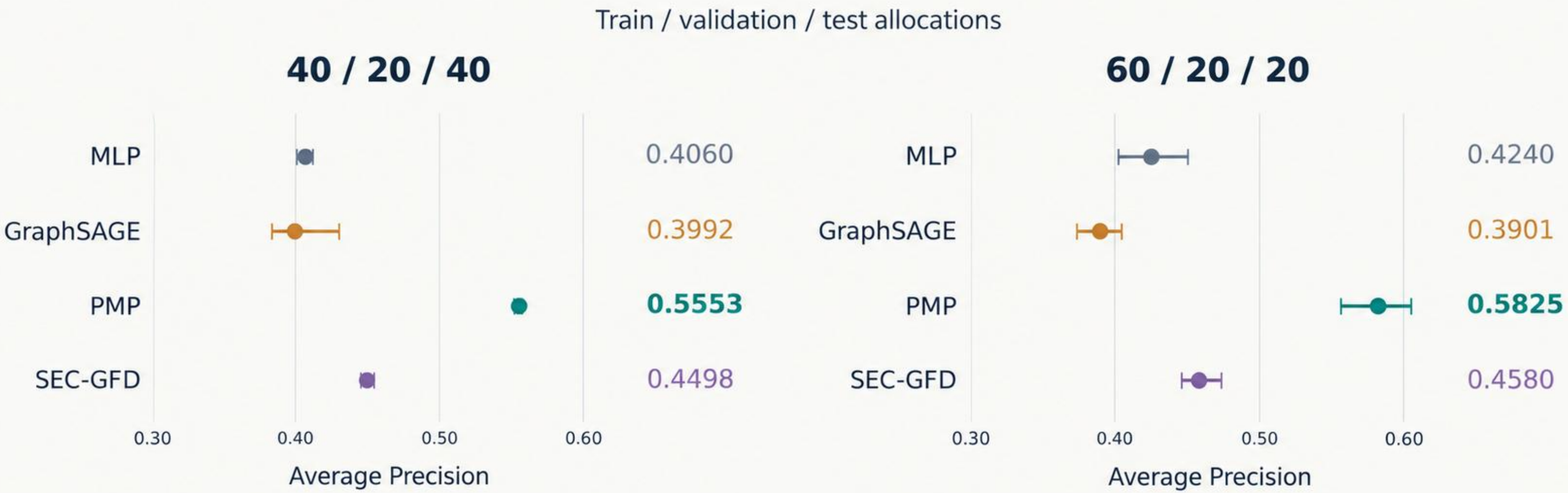
Non-oracle rewiring creates strong neighborhood churn but weak achieved label mixing.

Oracle rewiring uses true labels, including test labels, and remains a diagnostic.

Source: final report §4.2 · variant_audit.csv

PMP leads on the clean graph in both allocations

Average Precision · Clean graph · train_on_variant



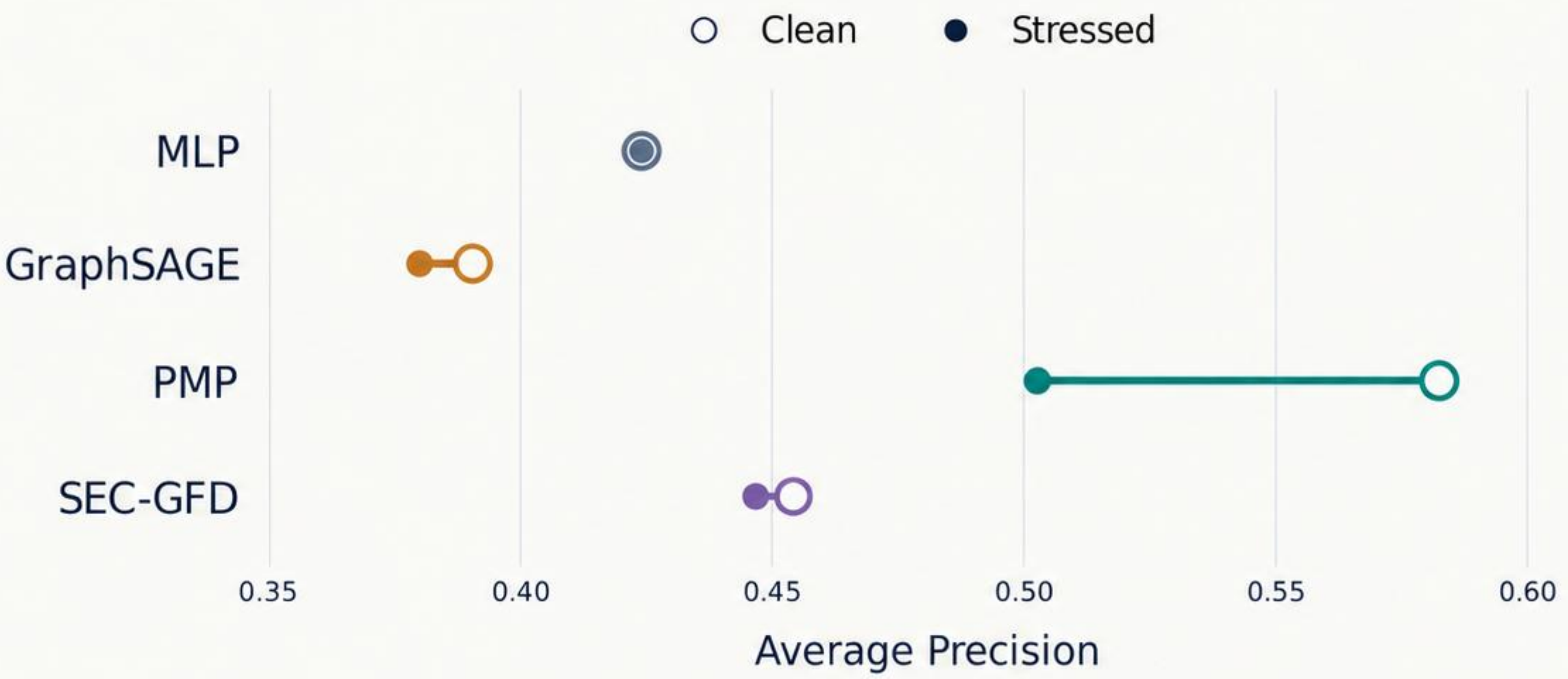
Points: cross-split means. Whiskers: descriptive 95% split-bootstrap intervals (3 splits).

PMP ranks first and SEC-GFD second across all three clean metrics.

MLP versus GraphSAGE depends on training-seed behavior; allocation changes also change the test cohort.

PMP loses more AP, yet remains strongest after shift

60 / 20 / 20 allocation · Clean-trained shift · Non-oracle rewiring at 0.30



	Clean AP	Stressed AP	AP loss
MLP	0.4240	0.4240	0.0000
GraphSAGE	0.3903	0.3804	0.0100
PMP	**0.5825**	**0.5022**	**0.0803**
SEC-GFD	0.4540	0.4467	0.0073

Endpoint: performance after shift | **Distance:** performance lost

The highest stressed score and the smallest drop are different robustness criteria.

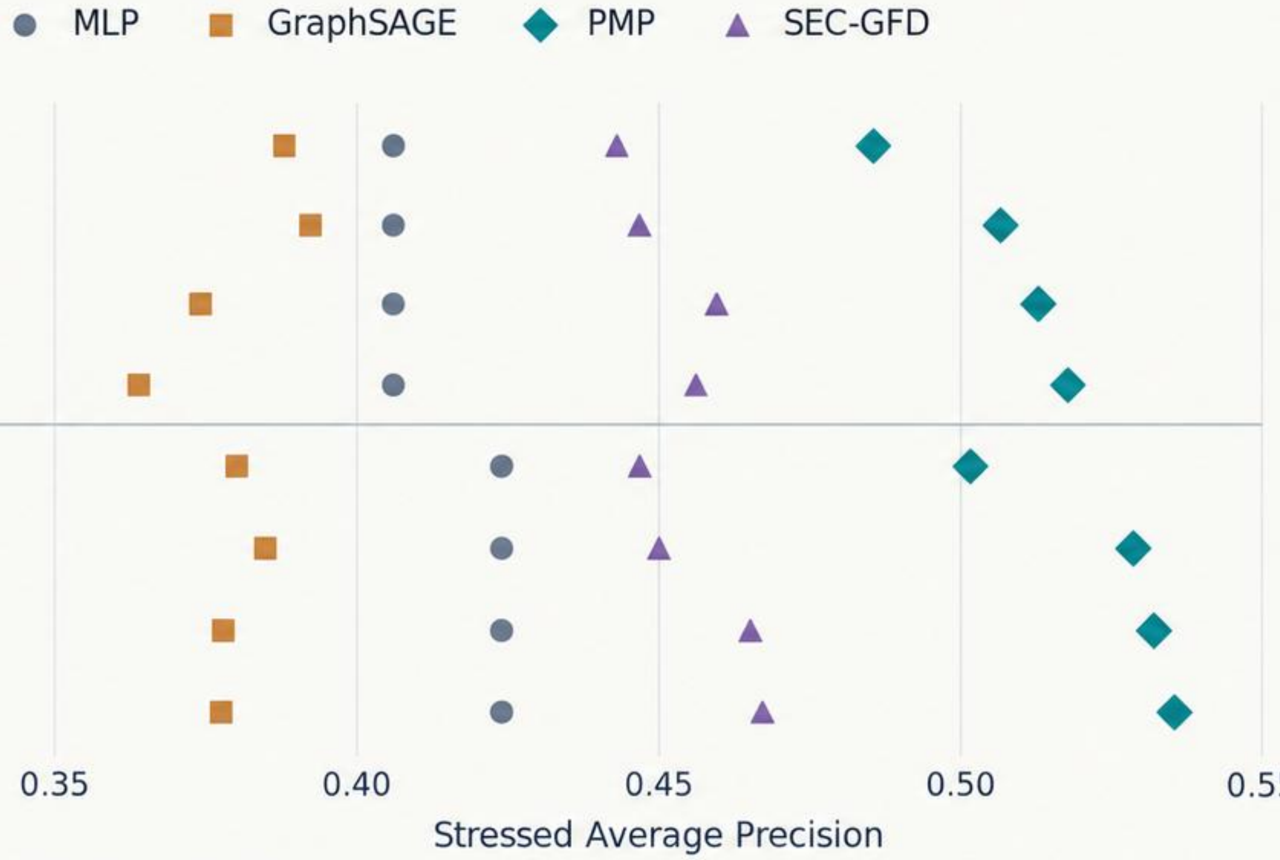
MLP ignores edges. Clean references match this protocol; values are rounded.

Source: final report §4.4 · performance_drop_max_stress_cross_split.csv

OPERATIONAL RESULTS ACROSS CONDITIONS

PMP’s operational lead survives both stresses and allocations

Maximum tested severity · Non-oracle scenarios only



Allocation	Protocol	Stress	PMP AP
40/20/40	Shift	Rewiring 0.30	0.4849
40/20/40	Shift	Noise 0.20	0.5062
40/20/40	Variant	Rewiring 0.30	0.5127
40/20/40	Variant	Noise 0.20	0.5175
60/20/20	Shift	Rewiring 0.30	0.5022
60/20/20	Shift	Noise 0.20	0.5287
60/20/20	Variant	Rewiring 0.30	0.5321
60/20/20	Variant	Noise 0.20	0.5355

Shift: train clean, evaluate changed graph. Variant: train and evaluate on the changed graph.

PMP leads all eight AP comparisons. The same winner pattern holds for ROC-AUC and macro-F1.

Consistent across the three tested splits; no claim of universal superiority.

Source: final report §4.4 · performance_drop_max_stress_cross_split.csv

Retraining helps PMP and SEC-GFD, but not every model

60 / 20 / 20 allocation · Non-oracle rewiring at 0.30

Positive contrast means higher AP after training on the variant.

PMP also gains +0.0278 AP in the 40 / 20 / 40 allocation.

Exposure to the changed graph helps some training recipes; it is not a universal remedy.

Variant training uses a fresh fit, not online recovery or fine-tuning of the clean model.

Feature camouflage harms every model

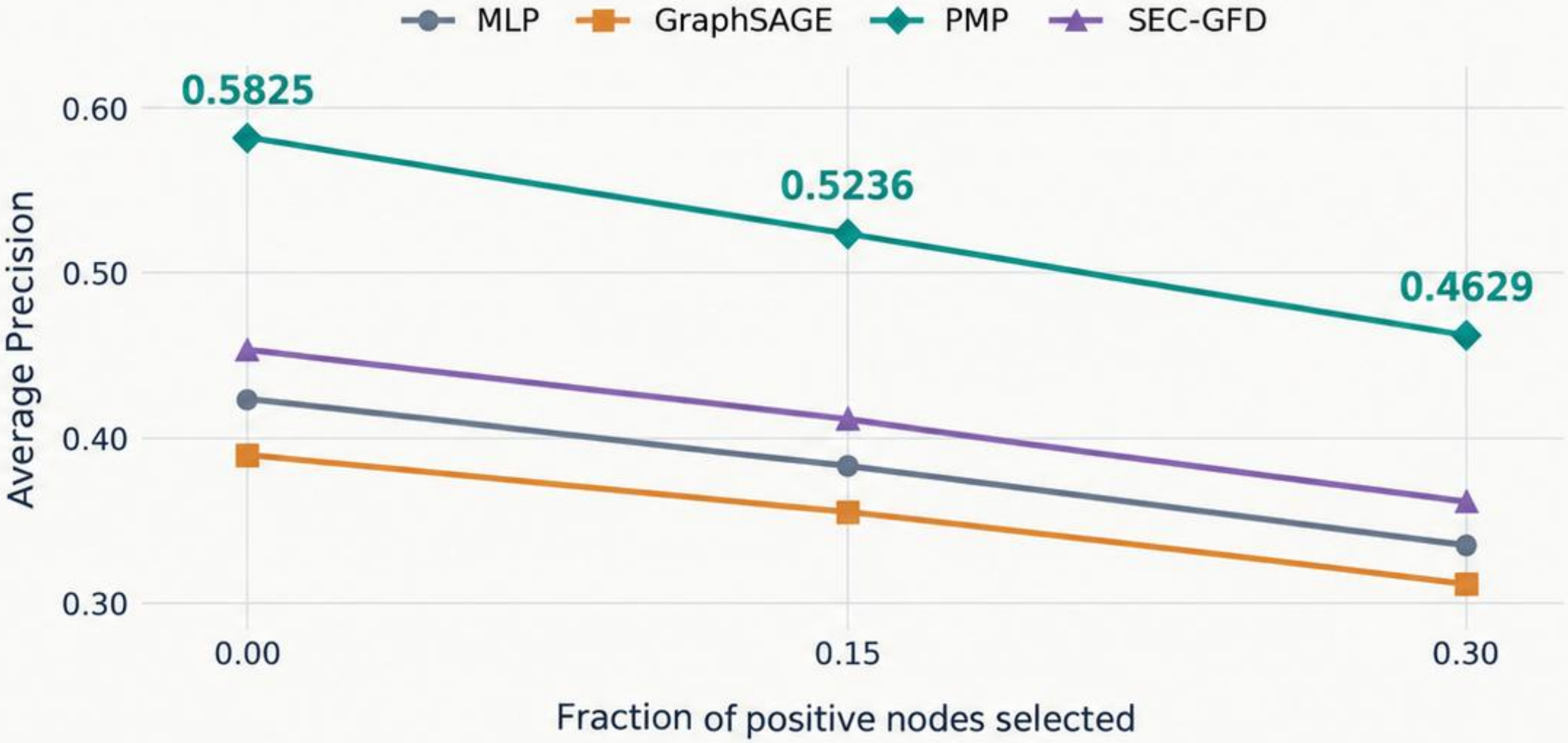
Oracle-assisted diagnostic

60 / 20 / 20 allocation · Clean-trained shift

Feature copying procedure

2,003 positive nodes
selected at severity 0.30

True labels guide target and donor selection.



Each selected node receives all 32 donor features. Its class label stays unchanged.

Graph context does not fully compensate for corrupted features; PMP still has the highest stressed AP.

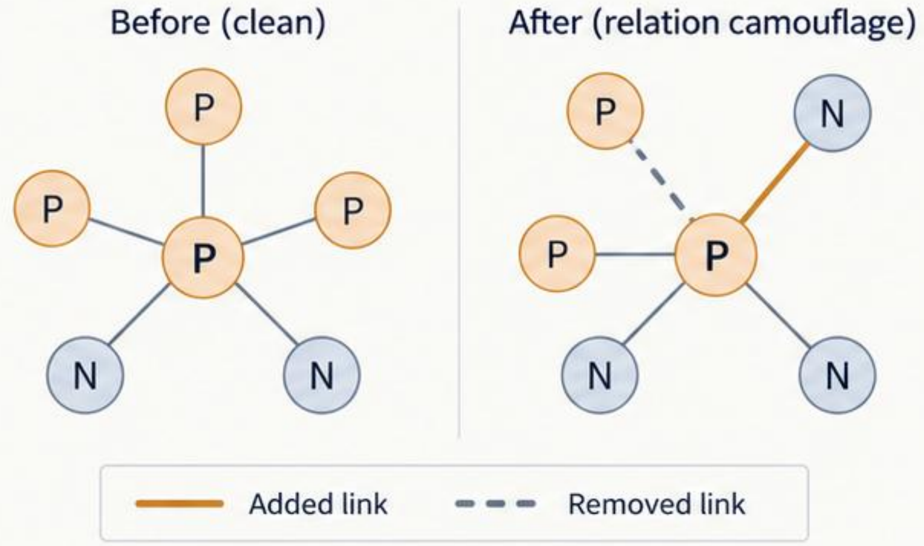
Only the plotted severities were evaluated. Label-assisted construction includes test-node labels.

Oracle-assisted diagnostic

This bounded relation camouflage had a modest effect

60 / 20 / 20 allocation · Clean-trained shift · Severity 0.30

Selected positive-node neighborhoods

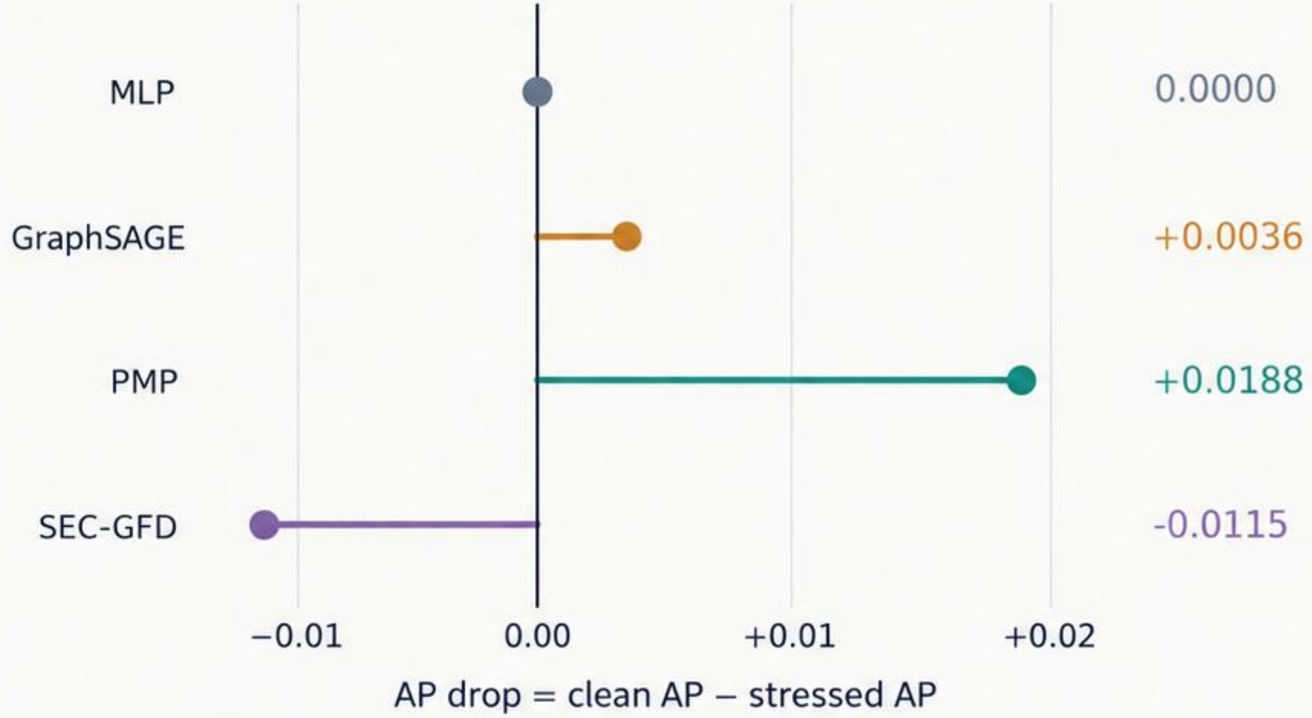


≈77,000 edges added **≈38,700** edges removed

At maximum tested severity

This is a schematic illustration; no exact counts are implied.

AP drop by model under relation camouflage



Positive = loss. Negative = improvement.

A modest effect under this treatment does not establish general camouflage resistance.

Construction uses true labels and changes degree and neighborhood composition together; it is not model-adaptive.

Oracle topology reverses SEC-GFD's result

Oracle-assisted diagnostic

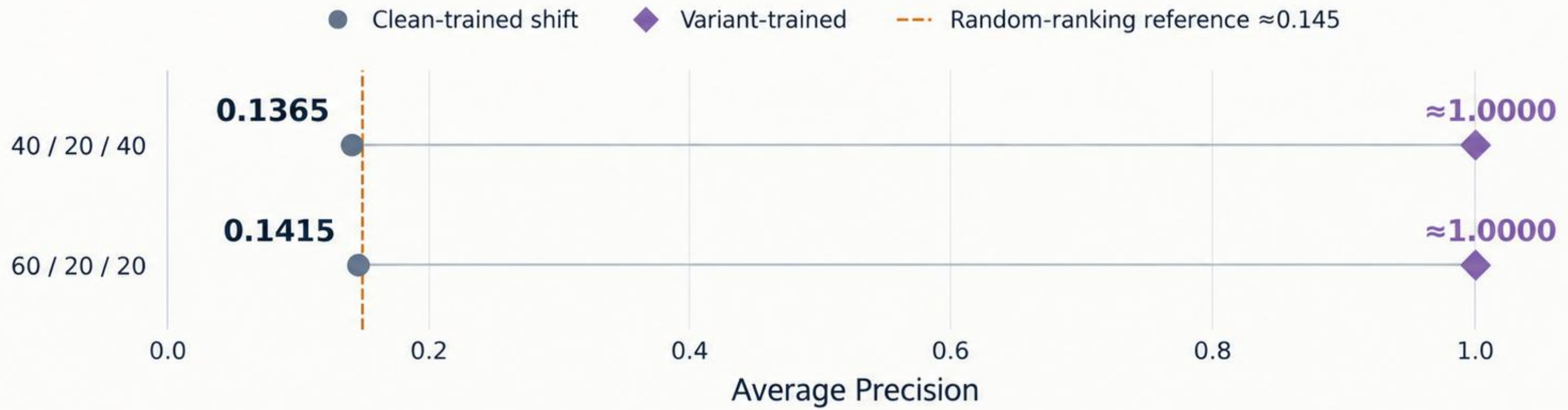
SEC-GFD adapter · Oracle rewiring at severity 0.30

Same oracle-constructed evaluation graph

Train clean, then face the shift

Train directly on the variant

Topology constructed using all true labels, including test labels.

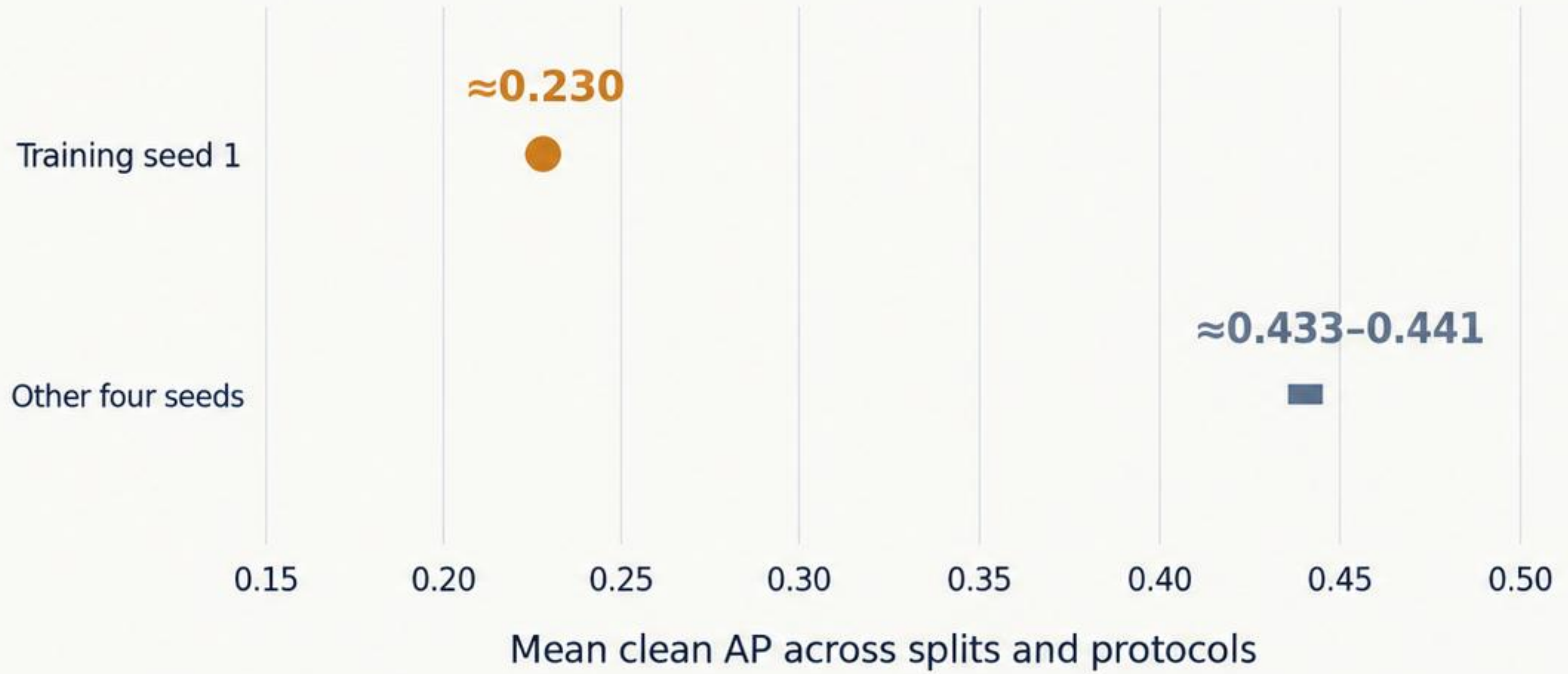


Near-perfect ranking after oracle training is a privileged structural signal, not operational robustness.

Excluded from the operational ranking. Aggregate values are rounded; not every run was exactly perfect.

One *prespecified* seed changes the baseline ranking

GraphSAGE on the clean graph



Segment: range of the other four seed means, not a confidence interval.

All five seeds retained

1. PMP
2. SEC-GFD
3. MLP
4. GraphSAGE

Sensitivity check: omit seed 1

1. PMP
2. SEC-GFD
3. **GraphSAGE**
4. **MLP**

Seed 1 remains in reported results

Keeping prespecified failures reveals training reliability that a favorable run would hide.

This is a finding about the executed training recipe; the failure mechanism was not isolated.

The reported results have a traceable evidence chain

Graph variants

Shared masks and cached interventions

`graph_variants.csv`

→

Realized-change audits

Requested versus achieved changes

`variant_audit.csv`

→

Unified results

Unique run keys and recorded metrics

`results.csv`

→

Derived evidence

Summaries, contrasts, and figures

`plots/*.csv`

Run keys connect each evaluated variänt to its audit and reported metrics.

5,040
unique successful result rows

0
missing or error rows

49
checksum comparisons passed

Each reported score can be traced to an evaluated *intervention* and retained evidence.
Checksums verify retained-artifact consistency. Deleted graph binaries must be regenerated.

Source: final report §§3.1, 3.8, 4.1, 4.12

The conclusions are strong within a bounded experiment

What the benchmark supports—and what the next study should test

Current boundary		Most relevant extension
One static, transductive YelpChi dataset	→	Additional datasets and temporal or inductive evaluation
Homogeneous adapters and unequal tuning recipes	→	Native relation-aware comparisons and matched tuning budgets
Limited splits, graph seeds, and severity points	→	More independent repetitions and richer perturbation paths
Weak achieved non-oracle label mixing	→	Calibrated interventions separating label mixing from degree effects

The findings apply to the executed benchmark; broader claims need broader evidence.

Three-split intervals are descriptive. Synthetic stresses are non-adaptive. Allocation changes also change the test cohort.

Source: final report §4.12

Robustness needs the score, protocol, intervention, and variation.

PMP achieves the highest stressed performance in the non-oracle comparisons, despite larger losses under unexpected graph shift.

Within the executed homogeneous YelpChi benchmark.

- Absolute score and performance drop answer different questions.
- Feature camouflage harms every model.
- Oracle topology reveals a privileged protocol interaction.
- Audits and repeated seeds change the interpretation.

Score | Protocol | Realized change | Seed variation

The contribution is a controlled, audited benchmark that makes these distinctions visible.

AP, ROC-AUC, and macro-F1 answer different questions.

Average Precision (AP)

Primary reporting metric

Summarizes positive-class precision–recall ranking.

Useful when positive reviews are a minority.

1
2
3

Ranking quality

ROC-AUC

Early-stopping monitor

Measures how well scores rank positives above negatives.

Threshold-free; monitors validation performance

Higher rank (positive review)

Lower rank (negative review)

Macro-F1

Threshold-dependent metric

Averages the positive- and negative-class F1 scores equally.

Each class F1 balances precision and recall.

P

N

Equal class weight

Validation determines selection

Validation ROC-AUC	→	Select stopping point
Validation macro-F1	→	Select classification threshold

The selected threshold is frozen for test evaluation.

AP leads the analysis. ROC-AUC and macro-F1 provide complementary evidence.

Under clean-trained shift, the clean validation threshold is reused on every variant.

The benchmark compares implemented training recipes

Selected settings from the final executed run

Setting	MLP	GraphSAGE	PMP adapter	SEC-GFD adapter
Structure	32–128–128–2	2 mean layers; hidden 64	1 layer; hidden 48	Hidden 32; polynomial order 2
Dropout	0.5	0.5	0	Not listed here
Learning rate	0.001	0.01	0.01	0.01
Weight decay	0.0005	0.0005	0	0
Training loss	Weighted cross-entropy	Weighted cross-entropy	Unweighted cross-entropy	Weighted cross-entropy + environment/NCE term
Max epochs / patience	100 / 10	100 / 10	100 / 10	50 / 10

PMP feasibility settings

Fanout 10; batch size 512; row-normalized features.
Missing optional normalization classes replaced with identity modules.

SEC-GFD compatibility changes

Zero-in-degree convolution enabled.
Environment-loss indexing and numerical range corrected.

Shared evaluation does not isolate architecture from configuration and adapter choices.

All models use Adam. The integrations are not full reproductions of the original papers.

Source: final report §3.4 · Supports slides 4 and 18

Severity specifies different operations in each stress family

Zero severity is the clean reference

Scenario	Operation	Positive severities	True labels used?	Audit measure
Non-oracle rewiring	Redirect destinations across a feature-derived two-means partition	0.15, 0.30	No	Rewired count; true-label heterophily
Uniform edge noise	Add random directed edges in proportion to source relations	0.10, 0.20	No	Added-edge count; density
Oracle rewiring	Redirect destinations to opposite true-label nodes	0.15, 0.30	Yes	Rewired count; true-label heterophily
Feature camouflage	Fully replace selected positive-node features with negative-donor features	0.15, 0.30	Yes	Selected-node count; donor similarity
Relation camouflage	Add negative neighbors and remove some positive links of selected positives	0.15, 0.30	Yes	Edge changes; selected-neighbor composition

Rewiring severity: fraction of edge entries. **Camouflage severity:** fraction of positive nodes selected.
Noise severity: added edges relative to the clean edge count.

Feature and partition controls

Full feature replacement ($r_\gamma = 1$). Feature partition fixed per graph seed.

Relation budget

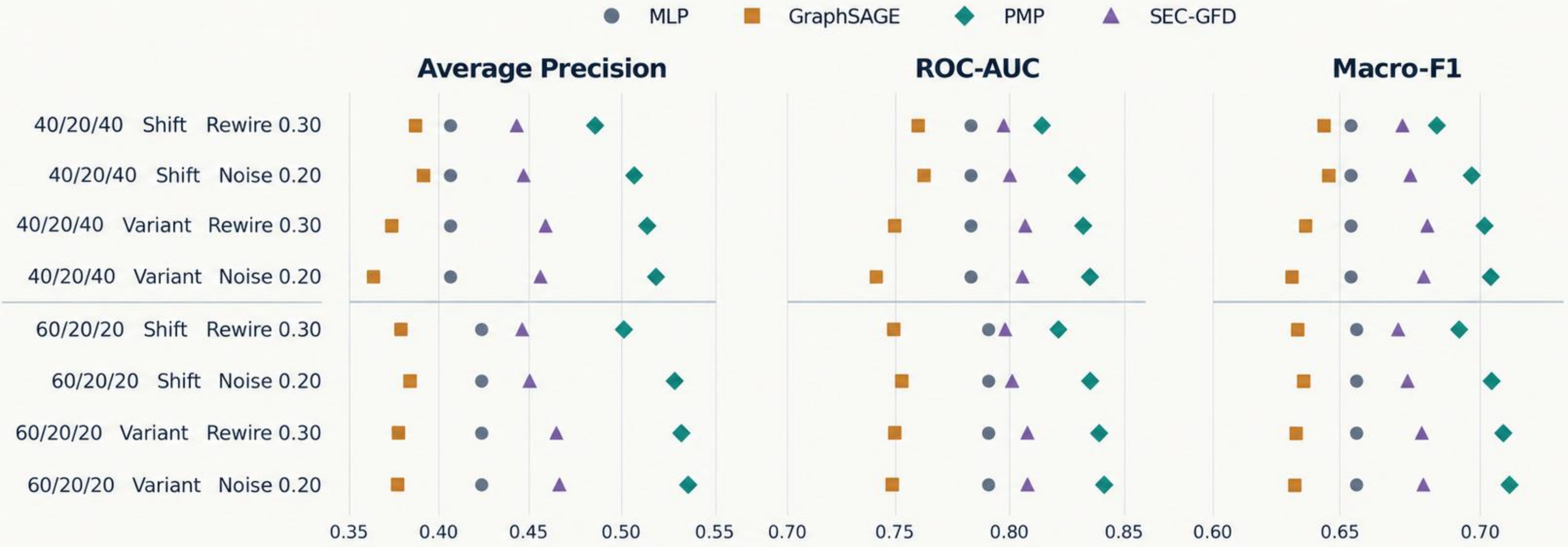
Add 25% of selected out-degree, bounded to 4–64 edges; remove half as many positive links.

Requested severity must be interpreted alongside the realized-change audit.

Oracle constructors may use test labels. These rows are excluded from operational ranking.

The operational winner pattern holds across all three metrics

Maximum tested severity · Non-oracle rewiring and uniform edge noise



Points are cross-split stressed means. Each metric panel uses its own labeled scale.
Shift: train clean, evaluate changed graph. Variant: train and evaluate on the changed graph.

PMP leads every displayed comparison; AP remains the primary metric.

Use matching protocol-specific clean references for drops. Uncertainty is descriptive, not a significance claim.

Source: final report §4.4 · performance_drop_max_stress_cross_split.csv

Evaluation rows are not independent experimental replications

How the counts are constructed

6 splits = 2 allocations × 3 split seeds

21 states per split = 1 clean + 5 families × 2 positive severities × 2 graph seeds

6 × 21 × 5 training seeds × 4 models × 2 protocols

= 5,040 evaluations

Variant-trained fits	2,520
Clean-trained fits reused under shift	120

Total: 2,640 model fits

What the uncertainty represents

Within a split

Graph and training seeds are crossed; clean-to-stress and protocol contrasts are paired.

Across splits

Allocation-level summaries bootstrap three split means with 1,000 resamples.

Shared graph plans

The same perturbation plans are reused across split masks.

Allocation sensitivity

40/20/40 and 60/20/20 change test membership and size, not only training-label count.

Three split means support descriptive intervals, not broad significance claims.

The main ordering survived the tested splits; this does not establish independence from split choice.

The retained evidence supports traceability and regeneration

Final evidence package: gfd-robustness-v4-factorial-report-r1

Evidence layer	Retained artifact	Purpose
Executed settings	active_config.json	Recover split-local settings
Variant construction	graph_variants.csv + variant_audit.csv	Identify interventions and realized changes
Model outcomes	results.csv	Inspect per-run metrics and status
Derived reporting	plots/*.csv	Recover curves, drops, and contrasts
Integrity records	Manifest + split receipts	Check retained-artifact consistency

Regeneration required

Downloaded dataset

Cached graph binaries

Isolated runtime

Upstream code clones

Reproducibility boundary

Seeded CUDA was not forced to be bitwise deterministic.

PMP samples neighborhoods during validation and testing.

Authoritative executed notebook

`KAGGLE_DUAL_T4_RESEARCH_RUN_with_outputs_latest_v4_r1_multi_seeds.ipynb`

Retained artifacts support logical traceability, not byte-for-byte checking of every executed graph.

Source: final report §§3.1, 3.8, 4.1, 4.12 · Supports slide 17

Prior work informs the models, stresses, and evaluation design

Executed graph methods

Inductive Representation Learning on Large Graphs
Hamilton, Ying & Leskovec · 2017 · NeurIPS
GraphSAGE: conventional message-passing baseline.

Partitioning Message Passing for Graph Fraud Detection
Zhuo et al. · 2024 · ICLR
PMP: partitioned message passing, integrated through an adapter.

Revisiting Graph-Based Fraud Detection in Sight of Heterophily and Spectrum
Xu et al. · 2024 · AAAI
SEC-GFD: spectral and environmental-constraint adapter.

Scenario and evaluation foundations

Enhancing Graph Neural Network-based Fraud Detectors against Camouflaged Fraudsters
Dou et al. · 2020 · CIKM
CARE-GNN: motivates feature and relation camouflage; not executed.

Label Information Enhanced Fraud Detection against Low Homophily in Graphs
Wang et al. · 2023 · The Web Conference
GAGA: related method and future comparator; not executed.

Pitfalls of Graph Neural Network Evaluation
Shchur et al. · 2018 · arXiv:1811.05868
Motivates repeated splits and training seeds.

Dataset context: Rayana & Akoglu (2015), Collective Opinion Spam Detection: Bridging Review Networks and Metadata · KDD. **Loader:** DGL FraudDataset documentation (1.1.3).

These papers motivate the benchmark; its scores are not direct reproductions of their leaderboards.